

ZHENG LU (ALICE)

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Data Scientist with 2+ years of experience building scalable ML and LLM-driven solutions with a focus on data pipelines and product analytics. Strong in translating business problems into insights through analytics and cross-functional collaboration. U.S. work authorization (OPT valid until Feb 2028); H-1B sponsorship required thereafter.

Experience

Verisk Analytics, Inc. | *Data Scientist* | *New Jersey, NJ* Feb 2025 – Present

Deliver small commercial business data for insurance carriers, enabling faster and more accurate underwriting and quote decisions through standardized attributes, with NAICS as a key signal.

- Defined and implemented endpoint segmentation logic to infer customer groups from API interaction patterns and measure feature adoption across cohorts. Developed **Sankey-based customer journey visualizations** to trace order flow and drop-offs, isolating key bottlenecks in the user journey and identifying product opportunities.
- Optimized processing of 10GB+ usage records using **Snowflake** and **PySpark**. Identified inconsistencies in login and query-handling logic; partnered with Engineering to deploy structured error monitoring and alerting, improving issue detection and reducing user-impacting defects by 30%.
- Redesigned a **database health monitoring dashboard** (DAX & Power Query) to monitor data coverage and attribute completeness, enabling effective data quality monitoring and prioritization of data enrichment efforts.
- Built a YAML-configured rule-based **text model** to flag the presence of services in business website content across **10+ industries**, supporting NAICS mapping and improving label consistency and coverage.

Verisk Analytics, Inc. | *Data Scientist Intern* | *New Jersey, NJ* Jun 2024 – Aug 2024

- Engineered a **semantic search pipeline** using BERT embeddings and cosine similarity to recommend NAICS codes from business descriptions. Fine-tuned Sentence-BERT on domain data to enhance semantic accuracy. Boosted classification coverage from 20% to 70% at 0.89 precision, enabling scalable, accurate NAICS assignments for long-tail business profiles.
- Developed **Generative AI** (GPT-4) with few-shot prompt engineering and logic-based self-validation to automate and scale NAICS classification. Achieved **0.95 precision** and **85% coverage**, scaling classification across 10x more records.

Allied Millennial Partners, LLC | *Financial Data Analyst Intern* | *New York, NY* Jun 2022 – Sept 2022

- Built **ETL pipelines (Apache Airflow, Databricks)** to process 10M+ financial records from multiple sources. Applied partitioning and data quality checks for missing values and outliers, reducing manual data cleaning effort by 60%.
- Queried and reconciled 1M+ statements using Snowflake SQL; extracted key indicators (NAV, P/E, volatility). Enhanced performance via optimized JOINS, refined CTE logic, and partition pruning—cut daily refresh time by 70%.
- Developed **time-series models (ARIMA, Prophet)** to predict asset returns. Increased accuracy by 7% through hyperparameter tuning and cross-validation. Integrated outputs into dashboards to support portfolio allocation decisions.

Huawei Technologies Co., Ltd | *Data Analyst Intern* | *Beijing* Apr 2021 – Aug 2021

- Partnered with marketing and data teams to analyze customer segments and behavioral patterns, supporting SEO strategy and keyword optimization that increased search visibility by 25%.
- Cleaned media event data in Excel and built Tableau dashboards for performance tracking and standardized reporting.

Skills

Data Analysis: SQL, Python (Pandas, NumPy), A/B Testing, Causal Inference, Experiment Design

Machine Learning: Naive Bayes, Regression (Linear, Logistic), Tree-Based Models (XGBoost, Random Forest), Clustering, Natural Language Processing, Time Series, Recommendation Systems

Data Engineering: Snowflake, BigQuery, PostgreSQL, AWS (EC2, S3, Athena), PySpark, Airflow

Product Analytics: KPI Definition, Funnel Analysis, Cohort Analysis, Dashboards (Tableau, Power BI), Google Analytics

Collaboration & Documentation: Agile/Scrum, Jira, Confluence, Git

Education

Columbia University *M.S. in Applied Analytics*

Sept 2023 – Dec 2024

University of California, Los Angeles (UCLA) *B.S. in Statistics, Data Science*

Sept 2019 – Jun 2023